

may also lie outside a human's audible range) or to those sounds having a certain minimum volume level.

Accelerometer: These are used to detect changes in inclination or speed of the robot.

Most of these sensors are available in the ready to use form in the market, but they are quite expensive because they offer very good precision. But for robotics enthusiasts, buying them is not always a good deal because you can construct majority of them on your own and those too offering reasonably good precision and efficiency.

2.8 Input and output methodologies

The changes sensed by the sensors are converted into voltage difference and this voltage difference is then sent to the “mind” of the robot, which is a microcontroller (or microprocessor). The robot's processing unit requires input electrical connections to receive the signals from sensors, as well as connections to outputs like a motor. These signals are in form of bits (BINary digiT*S*). Two types of interfaces are used to transfer these signals (or bits) from sensors to the robot's control computer or from robot's control computer to motors:

Parallel Interfacing

In a parallel interface, multiple bits of data are transferred simultaneously by eight separate wires. This type of data transfer offers high speed because more information can be transferred in less time. A typical parallel interface commonly used in day to day life is a printer port on a personal computer; it sends the data to the printer byte by byte (1byte=8 bits). Different characters including alphabets, numbers and symbols are represented by different combinations of the eight-bit data (as according to ASCII character codes).

Serial Interfacing

Though parallel interfacing offers more speed, they occupy a large number of input/output ports. This proves to be a serious disadvantage when you have limited number of input/output ports and can't expand their number.

Serial interfaces, on the other hand, send data on a single wire, and thus conserve on number of input/output ports used. They do this by sending each bit in a queue (one by one). Obviously, this process is much slower than the parallel data transmission. There is a variety of serial interface schemes available, using one, two, three, or even four input/output lines.